

Pre-work · The Brain and Brainstem

The Brain, and the Brainstem. Read both Part A chapters first. Almost every question on this material is a location question: name the region, then name what it does. Learn the map before you learn the list, because the map is what makes the list stick.

Module 5 · 2 of 5

Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1 Start with the reading.

Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

2 Organize what you read.

In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

3 Draw it.

In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4 Apply it.

Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5 Come back to it later in the week

, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

3 min

PANEL 1 OF 3

Turn the brain and brainstem chapters into mini visuals. Eight chunks. The hubs carry the regions, the grids carry the functional detail, and the chain is the development story that explains why the regions sit where they do.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 CHAIN

Neural tube to adult brain

One chain with arrows. Neural tube, then the three primary vesicles, then the five secondary vesicles, then the adult structure each one becomes. Write the ventricle that forms alongside each secondary vesicle, because the fluid spaces develop with the tissue around them.

2 HUB

The five cerebral lobes

One lateral view of a hemisphere at the hub. Spoke to frontal, parietal, temporal, occipital, and insula, and write the primary role of each at the end of its spoke. Draw the central sulcus, the lateral sulcus, and the parieto-occipital sulcus as the boundaries, and mark the insula as hidden inside the lateral sulcus.

3 GRID

The functional areas of the cortex

Area down the side, motor areas first, then sensory, then association. Lobe, exact gyrus or sulcus, and role across the top. Sixteen rows is a lot, so build it in three blocks and check each block by covering the location column.

4 LADDER

The three classes of cerebral white matter fiber

Three rungs, ordered by how far each fiber reaches: within one hemisphere, between the two hemispheres, and out of the cerebrum entirely. Write the class name and one named example beside each rung. The bottom rung carries the internal capsule, and that is the one the clinical questions are built on.

5 HUB

The deep brain in coronal section

The third ventricle at the hub. Spoke out to the thalamus with the massa intermedia, the hypothalamus below it, and the epithalamus with the pineal gland behind. Then move laterally to the caudate nucleus, putamen, and globus pallidus, and draw the internal capsule threading between them. Write the job of each structure beside it.

6 HUB

The cerebellum

The cerebellum at the hub. Spoke to the two hemispheres, the vermis between them, the folia of the surface, the cerebellar cortex, the arbor vitae, and the deep nuclei. Then draw the three pairs of peduncles reaching to the midbrain, the pons, and the medulla, and mark which pair carries output rather than input.

7 GRID

Midbrain, pons, and medulla compared

Region down the side, superior to inferior. Key structures, cranial nerve nuclei, and one thing that region alone does across the top. Star the medulla row, because the vital centers are the reason an injury there behaves differently from an injury anywhere else in the brain.

8 SPLIT

Anterior circulation against posterior circulation

One split. Internal carotid on one branch, vertebral and basilar on the other. Name the arteries that come off each and write the territory each one feeds. Then draw the three communicating arteries that join the two branches into a closed ring and label the ring.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

3 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

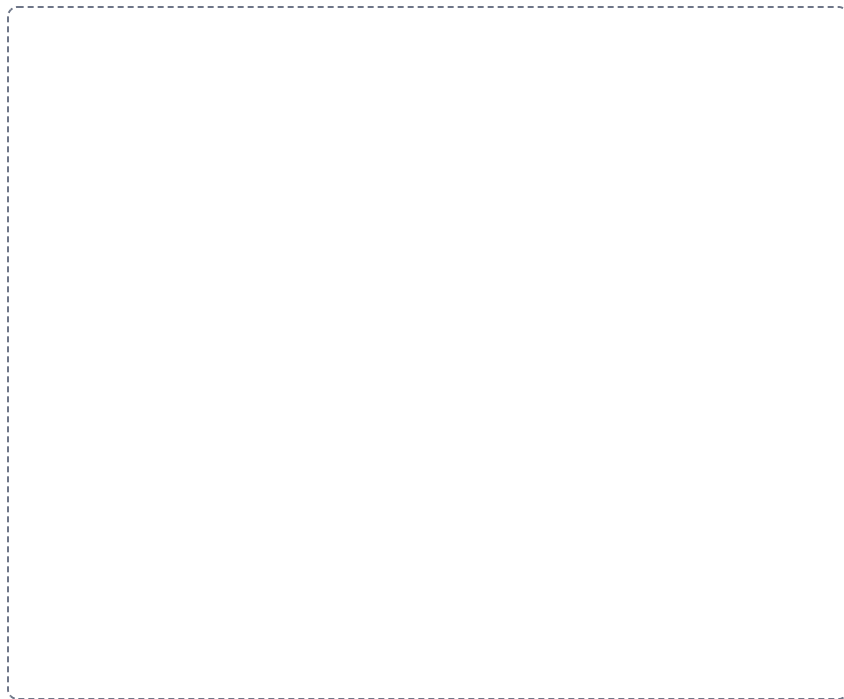
DRAWING 1

The lateral surface of the cerebrum, with the functional map on it

BRAIN DUMP FIRST

Two minutes. Five lobes, the sulci that bound them, and the named areas in position.

Draw one hemisphere from the side, gyri and sulci included. Draw the central sulcus, the lateral sulcus, and the parieto-occipital sulcus, then outline and label all five lobes, showing the insula peeled open inside the lateral sulcus. Now mark the primary motor cortex on the precentral gyrus, the primary somatosensory cortex on the postcentral gyrus, the primary visual, auditory, and gustatory areas, Broca's area, Wernicke's area, and the prefrontal cortex.



In your notebook, head the page **M5-7 The lateral surface of the cerebrum, with the functional map on it** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can be given any point on your own drawing and say which lobe it is in and what is lost if that spot is damaged.

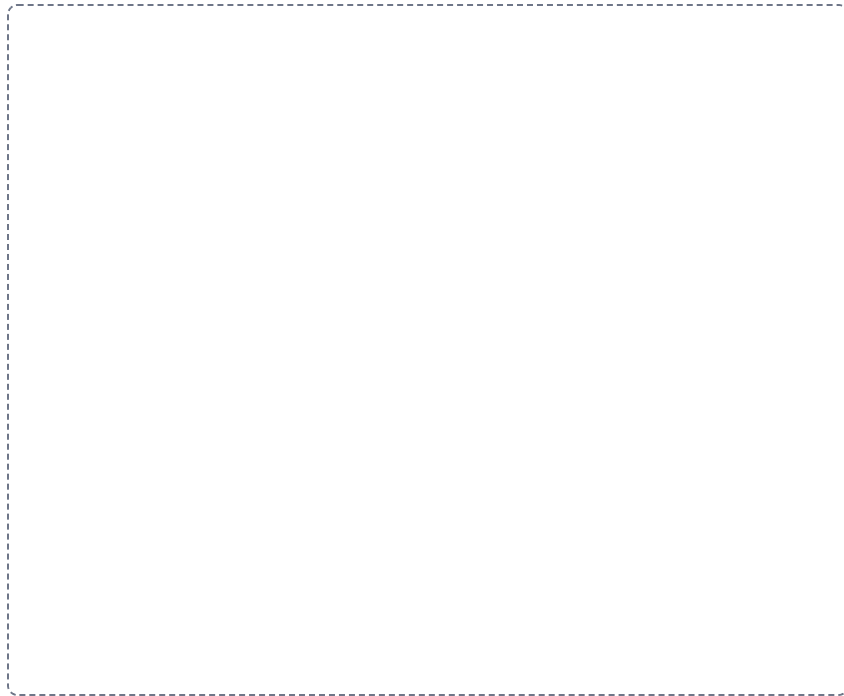
DRAWING 2

The brainstem, front and back

BRAIN DUMP FIRST

Two minutes. All three regions and every surface feature on each.

Draw the brainstem twice, once from the front and once from behind with the cerebellum removed. On the front view show the cerebral peduncles of the midbrain, the bulge of the pons, and the pyramids of the medulla with the decussation at the bottom. On the back view show the four colliculi of the tectum, the fourth ventricle, and the three pairs of cerebellar peduncles. Then write the cranial nerve nuclei held by each region beside it.



In your notebook, head the page **M5-8 The brainstem, front and back** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can name the region a cranial nerve nucleus sits in from the nerve number alone, and point to that region on your drawing.

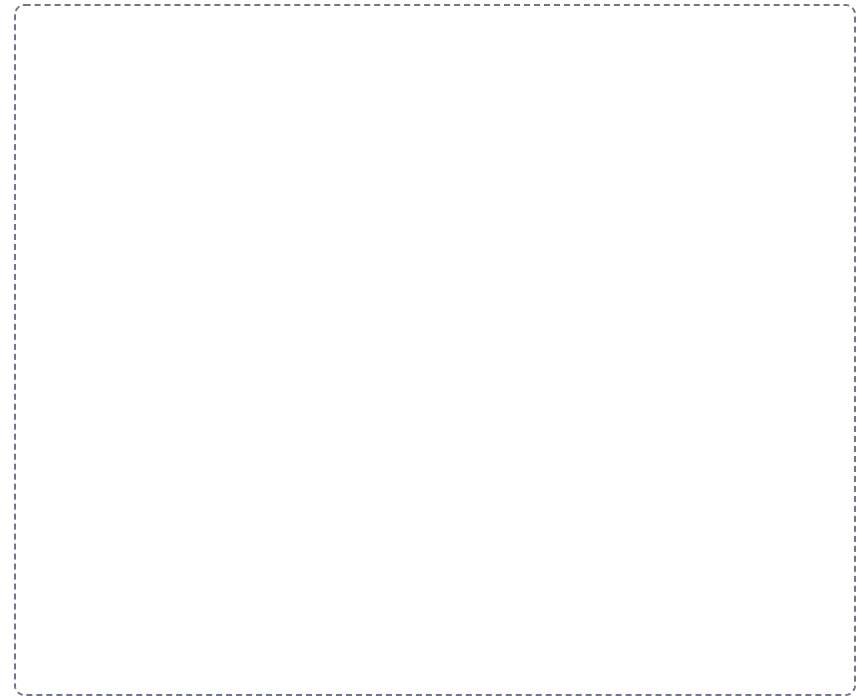
DRAWING 3

The brain in midsagittal section

BRAIN DUMP FIRST

Two minutes. Every region on the cut midline, cerebrum down to medulla.

Draw the brain sliced down the midline. Lay in the corpus callosum as the arch across the middle and the cingulate gyrus curving above it. Below the corpus callosum draw the thalamus, the hypothalamus, and the pineal gland of the epithalamus around the third ventricle. Continue down through the midbrain, pons, and medulla, and add the cerebellum behind them with the arbor vitae drawn inside it.



In your notebook, head the page **M5-9 The brain in midsagittal section** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can trace the four major parts of the brain in order on your own drawing and name the boundary between each pair.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A patient speaks in short, effortful phrases and struggles to get words out, but follows every spoken instruction correctly and can read. Name the area damaged, the gyrus it sits in, and the lobe.

Answer. Broca's area, in the inferior frontal gyrus of the frontal lobe, usually on the left.

Reasoning. Split language into its two halves before you name anything. Comprehension of spoken and written language belongs to Wernicke's area at the temporal and parietal junction, and comprehension is intact here, so that area is ruled out. What is lost is the motor production of speech, and motor areas of the cortex sit in the frontal lobe, which narrows it at once. Broca's area is the one frontal area named for speech production, and it lies in the inferior frontal gyrus, usually on the left. Notice that the arcuate fasciculus links these two areas, so a lesion in that association bundle would produce a different pattern again, and the pattern of what is spared is what tells the three apart.

Set A · Name it

1. The sulcus separating the primary motor cortex from the primary somatosensory cortex is the _____.
2. The commissural tract joining the two cerebral hemispheres is the _____.
3. The branching, tree-like white matter of the cerebellum is the _____.
4. The ridges on the front of the medulla formed by the descending motor tracts are the _____.
5. Name the five secondary brain vesicles and one adult structure formed by each.

Set B · Predict it

1. A small stroke destroys part of the internal capsule. Predict how much of the body is affected and explain why so small a lesion does so much.

2. The substantia nigra degenerates. Predict the kind of deficit this produces and name the system it belongs to by function.

3. An injury damages the ascending reticular network of the brainstem. Predict the effect on the patient and say why cortical damage does not do the same thing.

4. The posterior cerebral artery is occluded. Predict which lobe loses its supply and what function is lost.

Set C · Tell it apart

1. Distinguish the superior from the inferior colliculi by what each one serves and by what they do not do.

2. Explain why an injury to the medulla is immediately life-threatening while an injury to one cortical area is not.

3. A slow narrowing of one internal carotid artery causes fewer symptoms than a sudden blockage of the same vessel. Explain, using the ring at the base of the brain.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

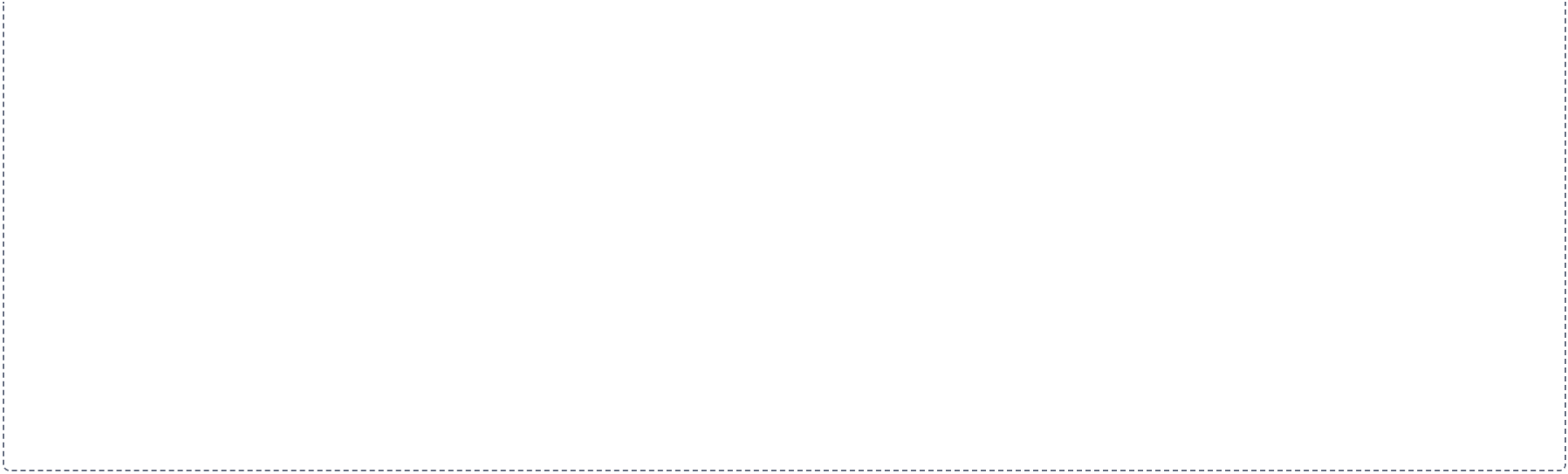
Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

Neural tube to adult brain

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.



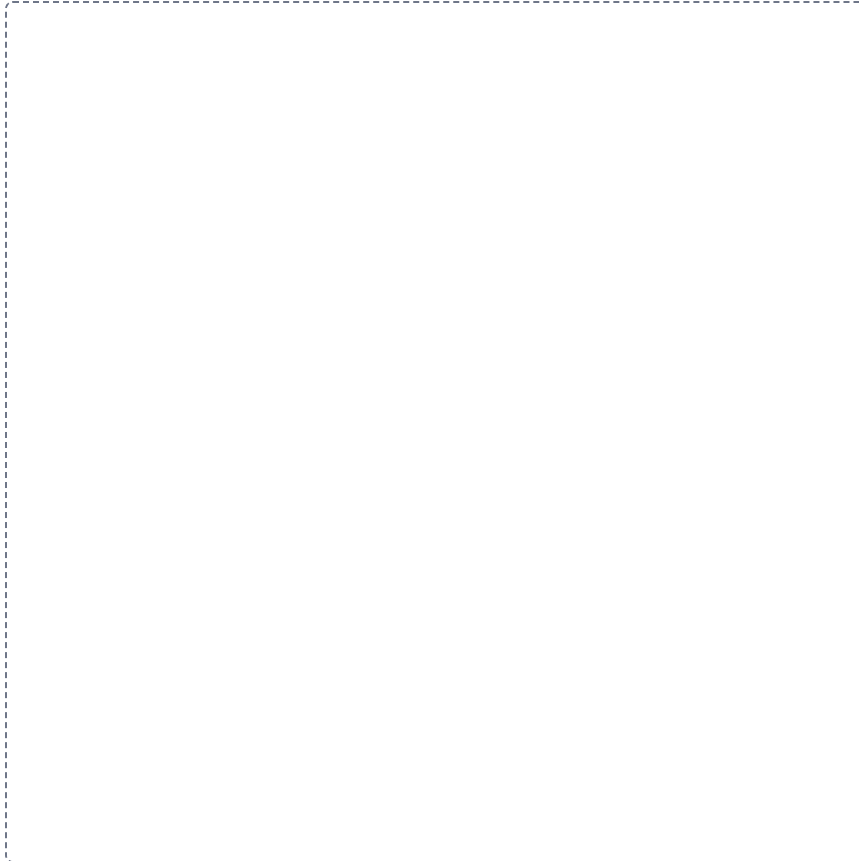


15-2

The five cerebral lobes

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

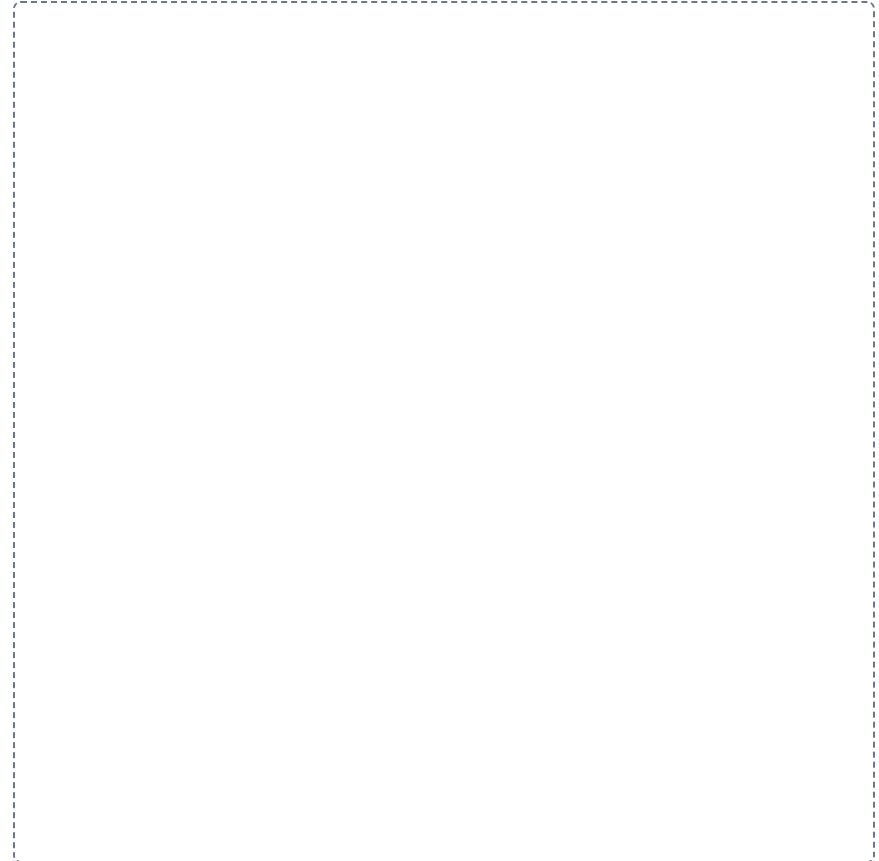


15-3

The functional areas of the cortex

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



15-2

The three classes of cerebral white matter fiber

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

15-3

The deep brain in coronal section

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

The cerebellum

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

Midbrain, pons, and medulla compared

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

Anterior circulation against posterior circulation**SPLIT**

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

